

# Leakage-free temporal protocol

**H History OD**

0-35% · model input

**Y Current labels**

35-70% · exact regional gain

**F Future test**

70-100% · frozen evaluation

Road graph  
+ historical OD

**Z0**  
parameter-free  
bidirectional diffusion

**BRIDGE (G4)**  
frozen Z0  
+ neural residual

Global ranking  
budget-aware head

Hard-disjoint  
region set

Spatially isolated  
candidate regions  
(Jaccard groups)

Selective CRP-style  
boundary overlay  
(existing exact substrate)

Exact bidirectional query  
+ local endpoint access  
(distance preserved)

*Learning allocates offline materialization resources; online shortest-path answers remain exact.*